

Task 2 – Attendance OCR Project (Flow Documentation)

Overview

This project automates the extraction of attendance data from uploaded images or PDFs using **OCR (Optical Character Recognition)**. It provides a simple frontend for users to upload files, and a backend that processes them, extracts data, and allows exports to CSV, Excel, or PDF.

System Architecture

1. **Frontend (React + Vite + TailwindCSS)**
 - Upload images/PDFs
 - Display extracted table data
 - Allow manual editing
 - Export table to **CSV / Excel / PDF**
2. **Backend (FastAPI + Python)**
 - Handles file uploads
 - Processes files using **Nutrient API (OCR)**
 - Extracts structured data (Date, Name, Check-in, Check-out)
 - Returns JSON response to frontend
3. **Deployment (Render)**
 - Backend: FastAPI app deployed as a **Web Service**
 - Frontend: React app deployed as a **Static Site**
 - Connected via API URL

Workflow

1. **Upload**
 - User uploads an attendance sheet (image/PDF) from frontend.
 - File is sent to the backend via `/upload` endpoint.
2. **Processing (OCR)**
 - Backend saves the file temporarily.
 - Sends it to Nutrient OCR API.
 - Extracts text → Converts into structured table.
3. **Display**
 - Frontend receives structured data.
 - Shows it in an editable table with **Serial No, Date, Name, Check-in, Check-out**.
4. **Export**
 - Users can export data to:
 - **CSV** → Raw comma-separated data.
 - **Excel (XLSX)** → Spreadsheet format.
 - **PDF** → Well-formatted attendance sheet with header + serial numbers.